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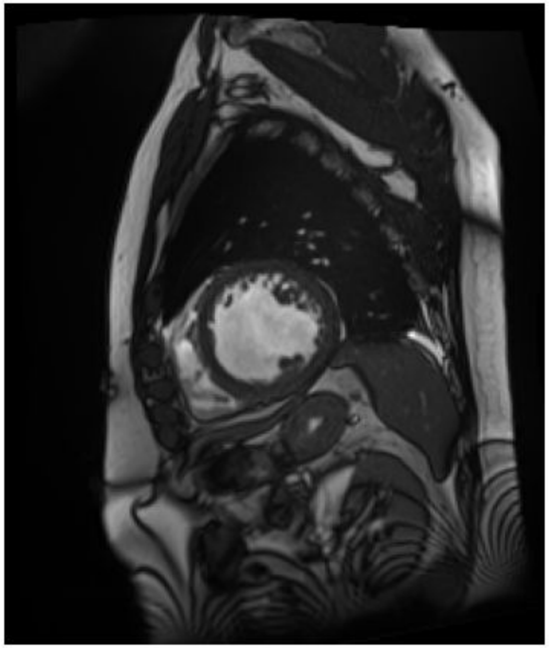
Implementing a Deep Learning Approach for Left Ventricle Segmentation from Cardiac MRI

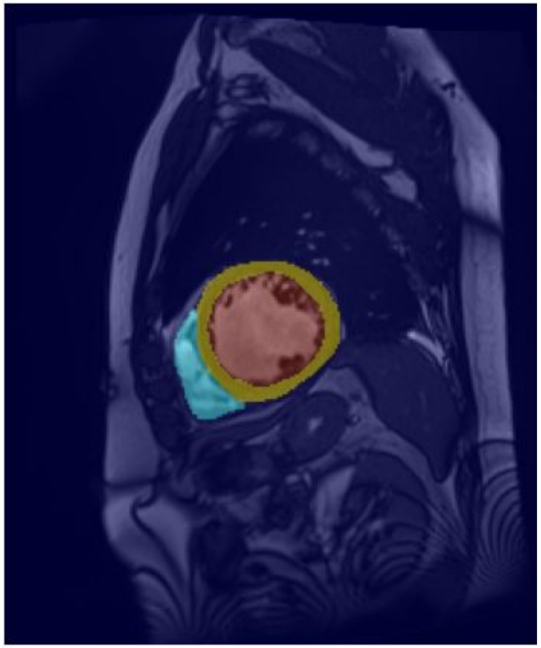
Based on: "A Deep Learning-Based Approach for Automatic Segmentation and Quantification of the Left Ventricle from Cardiac Cine MR Images”

Deep Learning S2026
Manuela Betancur

Motivation



Cardiac MRI.



- Red – Left ventricular cavity (LV cavity)
- Yellow – Left ventricular myocardium (LV myocardium)
- Light blue – Right ventricular cavity (RV cavity)

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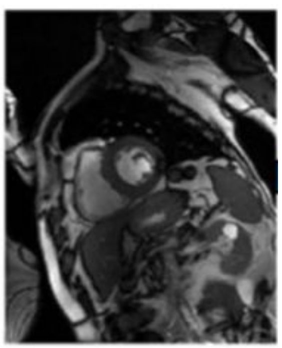
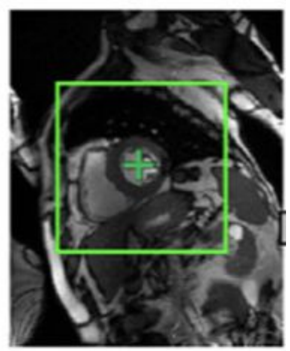
Proposed framework

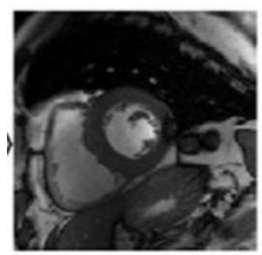


Cardiac MRI

Region of interest localization

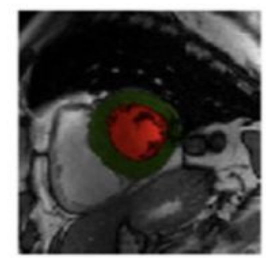


FCN 1:
Find the cavity



Crop the image to 128x128 taking the cavity center

Final segmentation

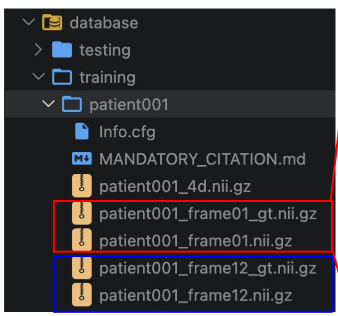


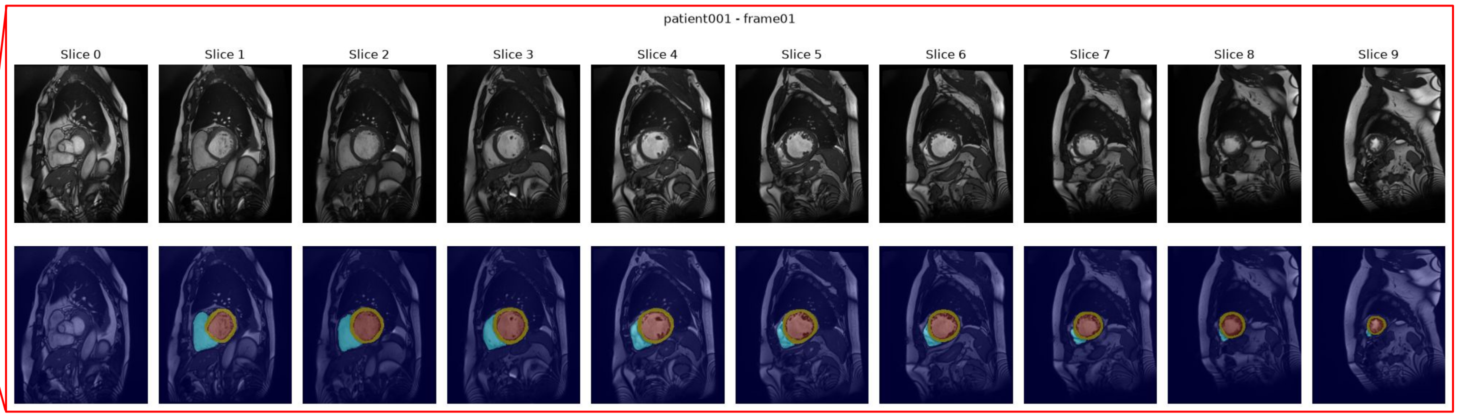
FCN 2:
Segmentation of cavity and myocardium

Dataset

ACDC-2017 dataset

- Training: 100 patients (90 used for training and 10 used for validation)
- Testing: 50 patients
- Each patient includes 2 frames (heart fully filled and heart fully contracted)
- 8–12 short-axis slices per frame each with a corresponding manual segmentation

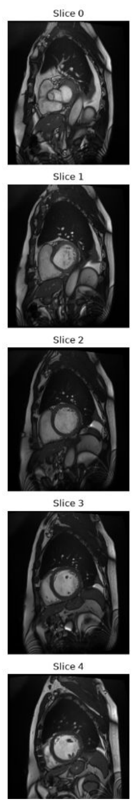




Locally-acquired dataset (not used in this approach)

- Used to evaluate the model's generalization
- 11 patients, 26 CMR scans (~6,000 images), 25 temporal frames per slice (full cardiac cycle).

FCN 1 - Image preprocessing pipeline

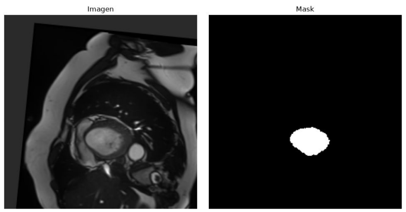


Resized images to 256 x 256

Applied transformations for data augmentation

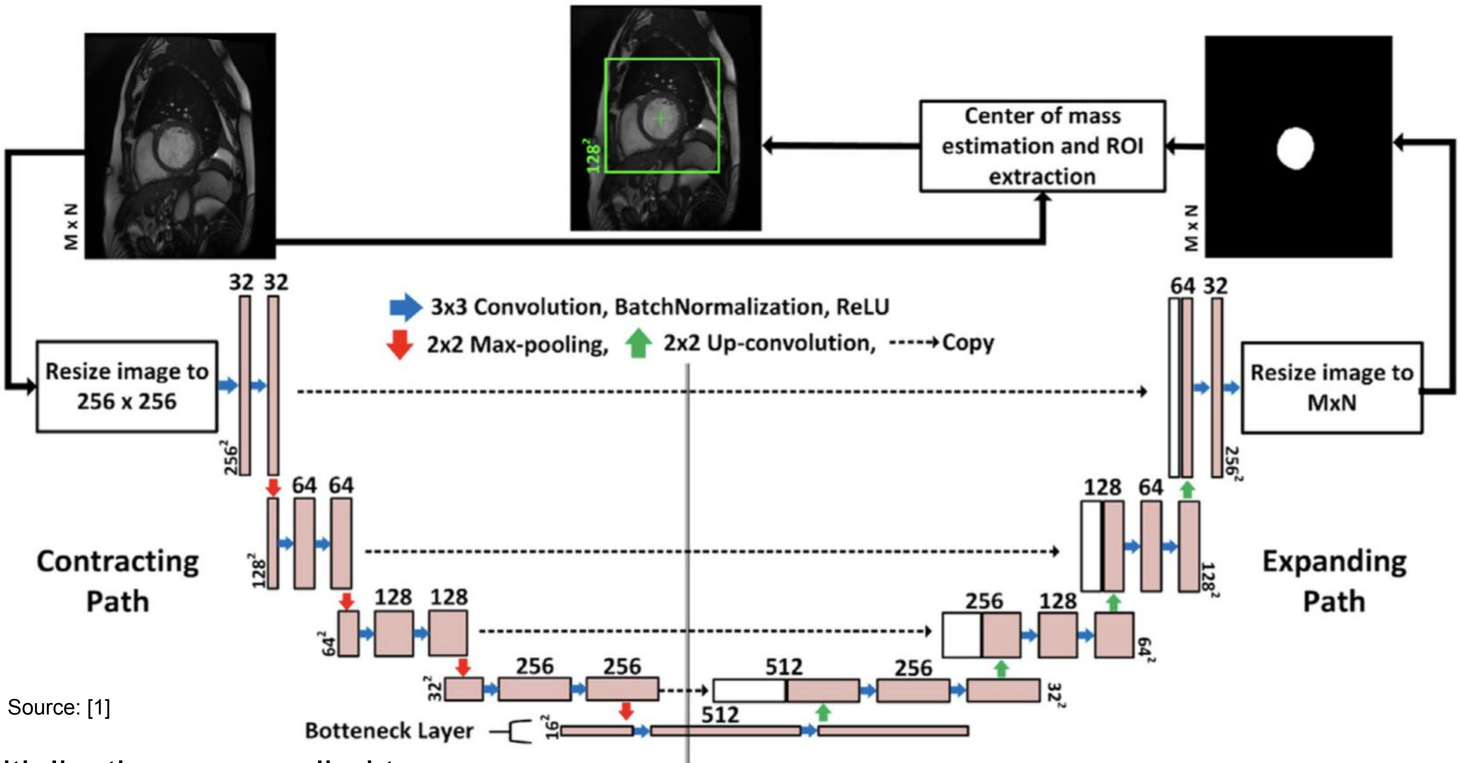
- Scaling (0.9 - 1.1)
- Translation (10%)
- Rotation (-10° to 10)

Only for the training set



Also, MRI images were normalized

FCN 1 architecture



Source: [1]

Kaiming initialization was applied to convolutional layers before training.

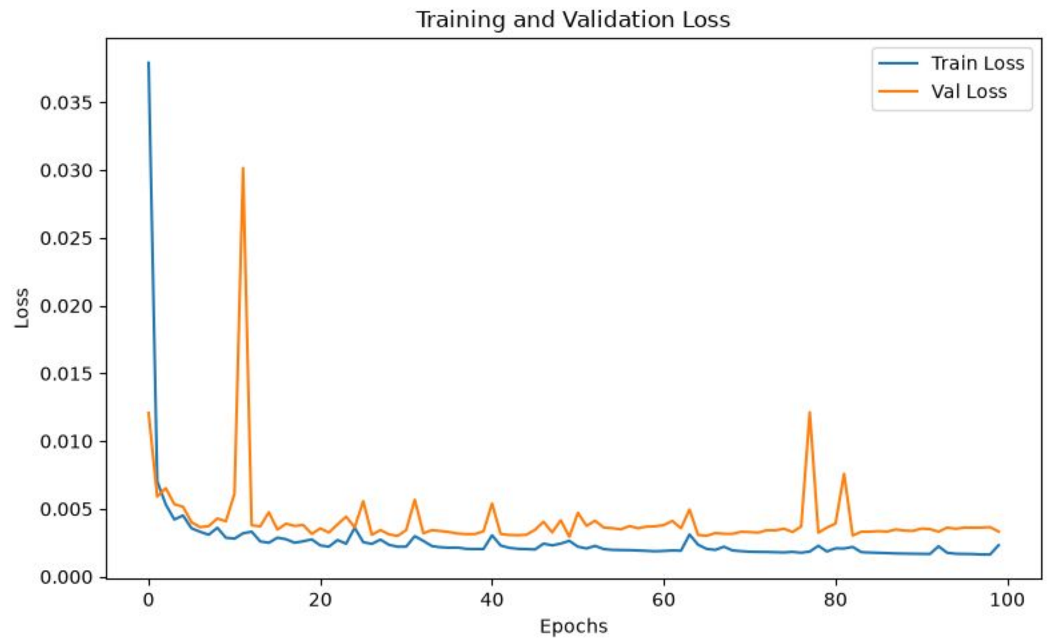
FCN 1: Training and validation

Hyperparameters

- Learning rate: 0.001
- Bach size: 8
- Epochs: 100
- Adam optimizer

Grid Search

- Learning rate
  [0.01, 0.001, 0.0001]
- Batch size
  [8, 16, 32]
- *Epochs 100:50:300 - Not taken in this approach*



FCN 1: Evaluation

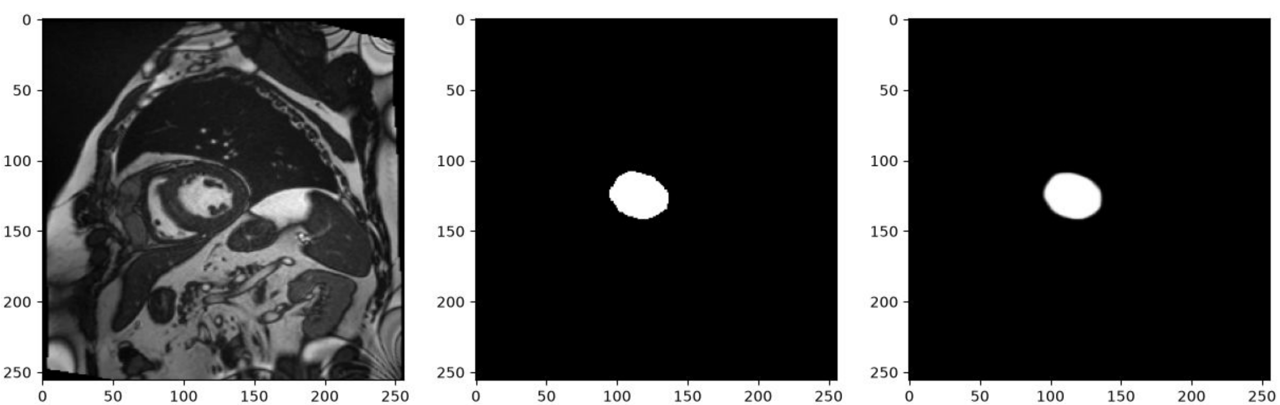
After generating the initial segmentation mask, the center of mass of the Left Ventricle (LV) cavity was calculated

The original paper used connected component analysis to clean the output by removing false-positive pixels (noise reduction) - *not implemented in our approach.*

```
Evaluation Complete
Mean Error: 1.56 ± 2.99 pixels
Min Error:  0.03 pixels
Max Error:  26.69 pixels
----------------------------------------
```

	Mean	STD	Max.
Hough transform (Khened et al., 2019)	4.00	3.83	36.24
Proposed (FCN1)	1.41	1.65	5.00

Source: [1]

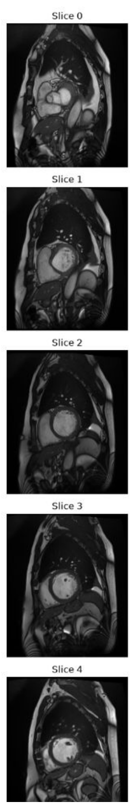


ground truth   predicted

One of the first attempts: Without data augmentation, 30 epochs

```
Evaluation Complete
Mean Error: 2.82 ± 7.25 pixels
Min Error:  0.01 pixels
Max Error:  106.27 pixels
---------------------------------
```

FCN 2 - Image preprocessing pipeline



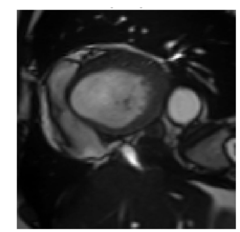
1. Resized images to 256 × 256

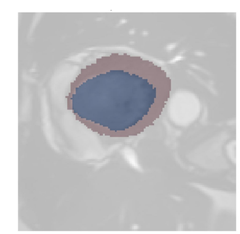
2. Applied data augmentation transformations:
- Scaling (0.9–1.1)
- Translation (10%)
- Rotation (−10° to 10°)

3. Found the true center of the cavity (x1,y1) and added random noise (xo,yo)
4. Cropped the image to 128 × 128, centered at (x1+xo, y1+yo)

Only for the training set

Also, MRI images were normalized





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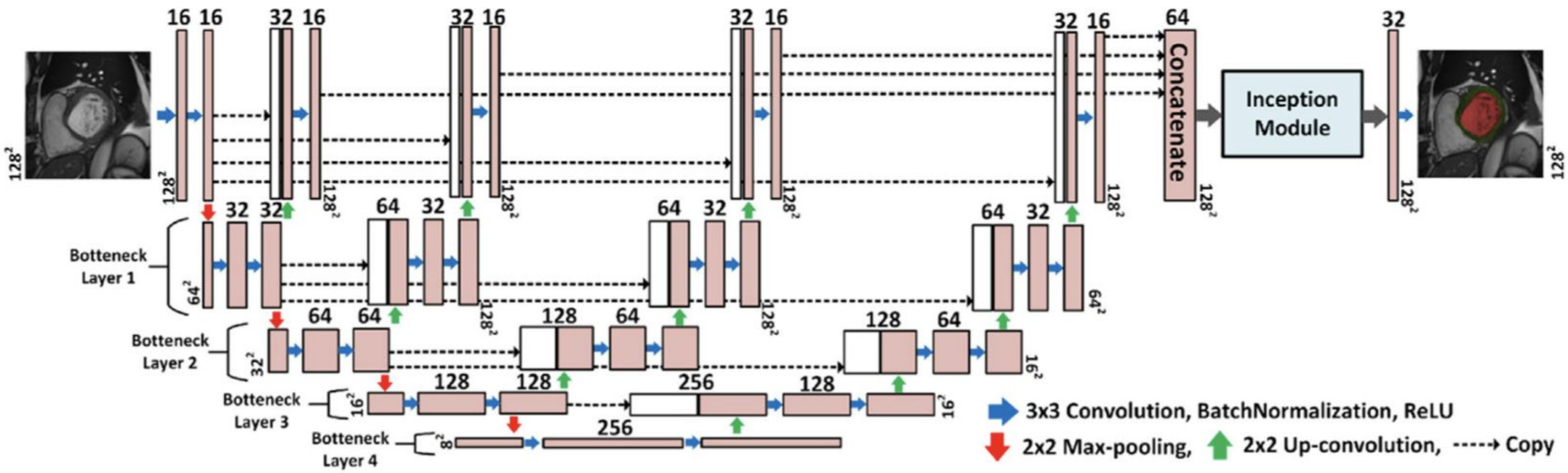
FCN 2 architecture

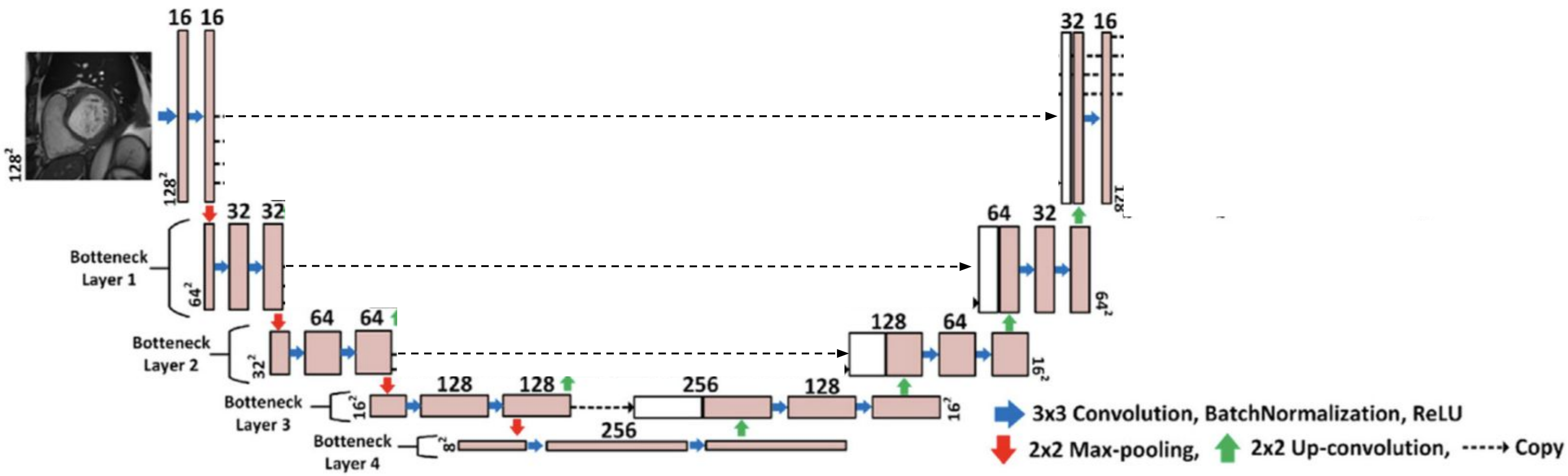
Source: [1]



Kaiming initialization was applied to convolutional layers before training.

FCN 2 architecture

Source: [1]



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FCN 2 architecture

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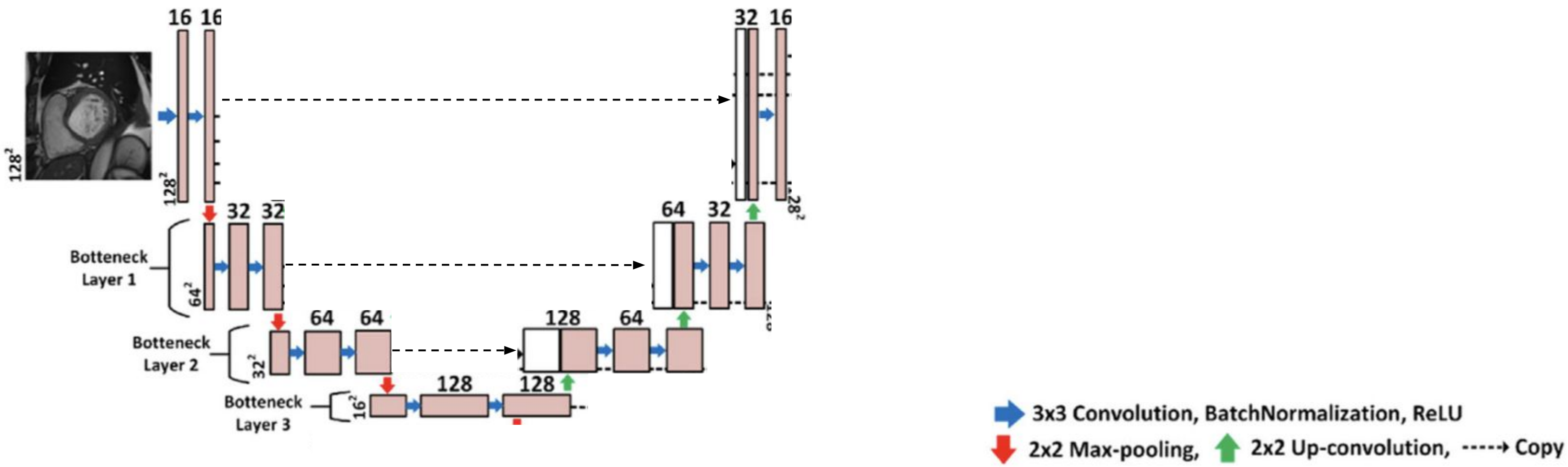

FCN 2 architecture

Source: [1]

FCN 2 architecture

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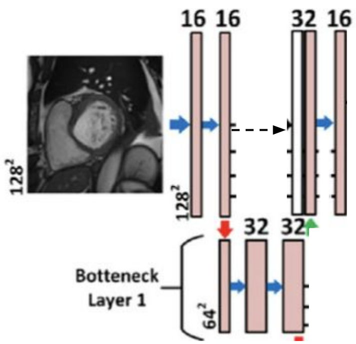

3x3 Convolution, BatchNormalization, ReLU

2x2 Max-pooling, 2x2 Up-convolution, ----→ Copy

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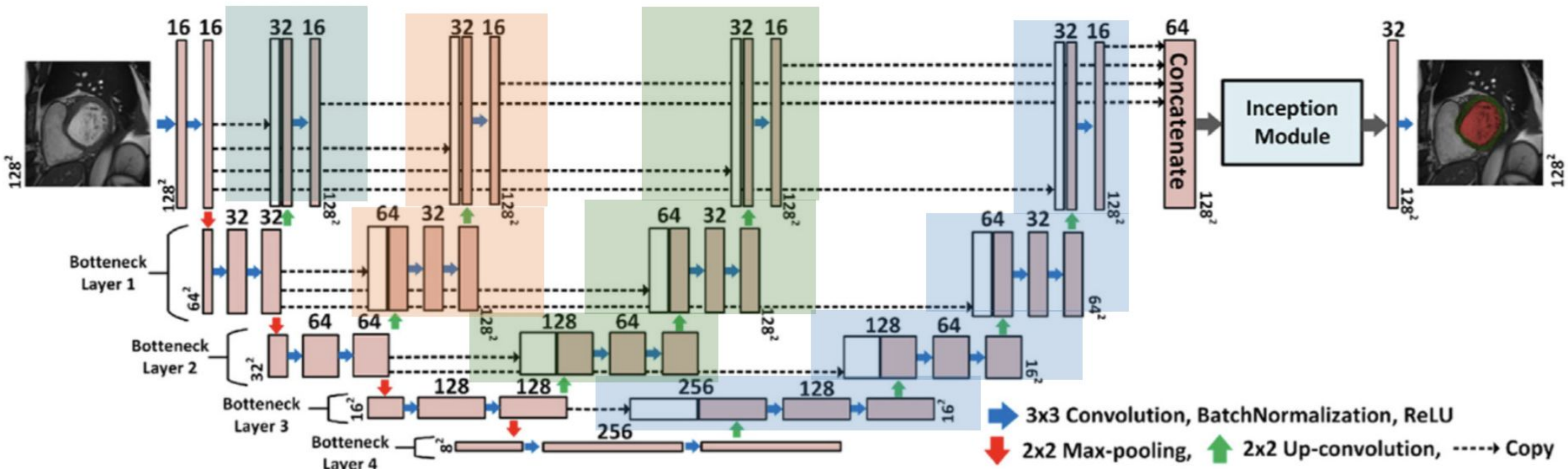
FCN 2 architecture



Source: [1]

Kaiming initialization was applied to convolutional layers before training.

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FCN 2: Training and validation loss

Hyperparameters

- Epochs: 100
- Learning rate: 0.001
- Bach size: 8
- Adam optimizer

Grid Search

- Learning rate
  [0.01, 0.001, 0.0001]
- Batch size
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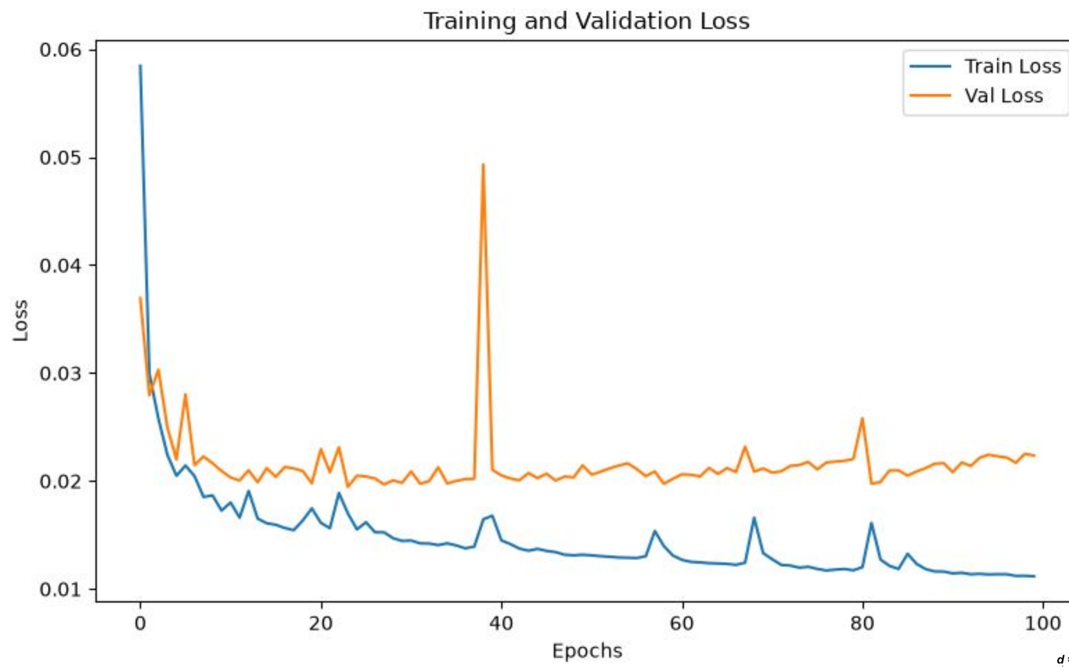
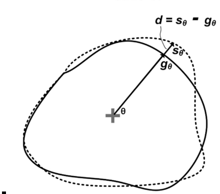
- *Epochs 100:50:300 - Not taken in this approach*



FCN2 Loss Function

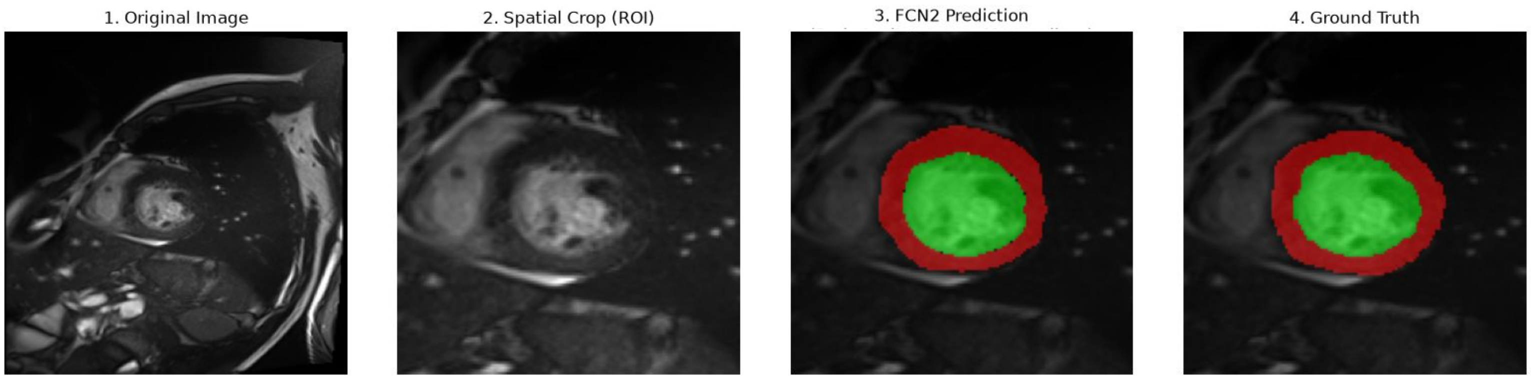
- Original Method: Cross-Entropy (CE) + Radial Distance (RD) loss.
- Simplified Approach: CE loss only (RD omitted).



Radial distance

True round border vs predicted round border

Full Pipeline Visualization: FCN 1 + FCN 2



Loss function	Dice coeff.	
	LV Cavity	MYO
L_{CE}	0.93	0.86
$L_{CE} + L_{RD}$	0.94	0.89

Source: [1]

In this approach, the Overall Average Dice Coefficient for the combined LV Cavity and Myocardium is 0.85 (Only LCE)

References

[1] H. Abdeltawab, F. Khalifa, F. Taher, N. S. Alghamdi, M. Ghazal, G. Beache, T. Mohamed, R. Keynton, and A. El-Baz, “A deep learning-based approach for automatic segmentation and quantification of the left ventricle from cardiac cine MR images,”